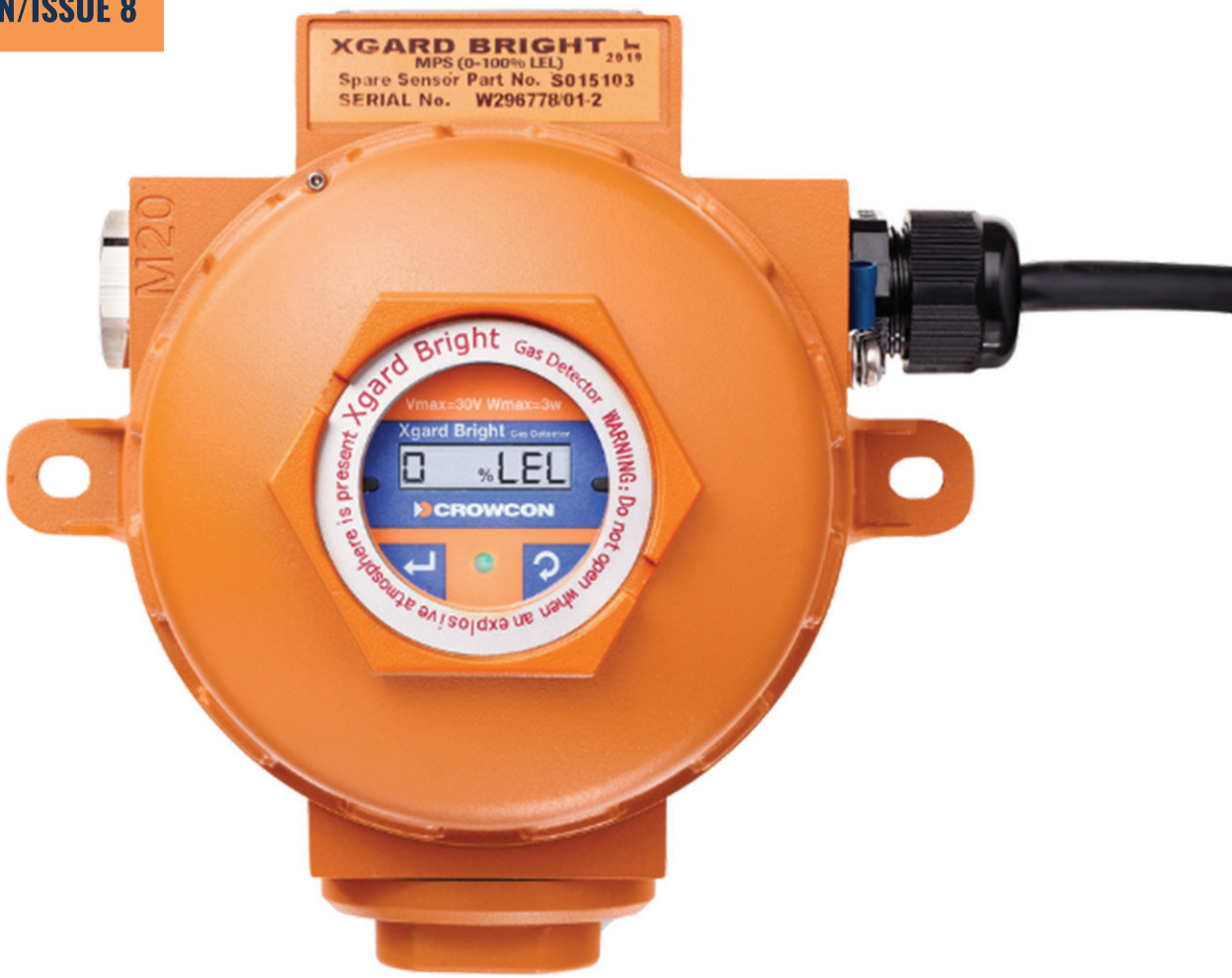
PRODUCT MANUAL

# XGARD **BRIGHT**

Addressable fixed point gas detector with display

May 2026

**M070068/EN/ISSUE 8**




crowcon
DETECTING GAS. **SAVING LIVES.**

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# 1. INTRODUCTION

## 1.1 PRODUCT OVERVIEW

The Xgard Bright is a versatile gas detector with a bright OLED display and magnetic wand for easy operation. It supports analogue 4-20mA and RS-485 Modbus signals, with optional HART interface, and includes relays for alarms or control systems. It can be equipped with various sensors, such as electrochemical, catalytic, MEMS, or infrared sensors.

## 1.2 SAFETY INFORMATION

- Xgard Bright gas detectors must be installed, operated and maintained in strict accordance with these instructions, warnings, label information, and within the limitations stated.
- Xgard Bright detectors are designed to detect gases or vapours in air. Xgard Bright detectors with catalytic or MEMS sensors are not suitable for use in inert or oxygen deficient atmospheres.
- Electrochemical cells used in toxic and oxygen versions of Xgard Bright contain small volumes of corrosive electrolyte. Care should be observed when replacing cells to ensure that the electrolyte does not come into contact with skin or eyes.
- Maintenance and calibration operations must only be performed by qualified service personnel.
- Only genuine Crowcon replacement parts must be used, substitute components may invalidate the certification and warranty of the detector.
- Xgard Bright detectors must be protected from extreme vibration, and direct sunlight in hot environments as this may cause the temperature of the detector to rise above its specified limits and cause premature failure. A sunshade is available for Xgard Bright.

NOTE: If the sinter is subjected to direct water jet the product must be bump tested with gas to ensure correct operation.

- This equipment must not be used in a Carbon Disulphide atmosphere.

WARNING: DO NOT OPEN WHEN AN EXPLOSIVE ATMOSPHERE IS PRESENT.

- The lid on Xgard Bright must be kept tightly closed until power to the detector is isolated otherwise ignition of a flammable atmosphere can occur. Before removing the cover for maintenance, ensure the surrounding atmosphere is free of flammable gases or vapours.
- The cable gland must be installed before use and must comply with the requirements of standards EN60079-0 and EN60079-1 with minimum IP66 ingress protection.
- External earthing should be considered and installed according to these instructions before use. For use in North America, the internal grounding terminal shall be used as the equipment grounding means and that the external terminal is only a supplemental bonding connection where local authorities permit or require such a connection.
- Only the cable/wires which are specified in the user manual can be used.

**ATEX/IECEX AND INMETRO CERTIFIED ALUMINIUM VERSION:**

- This detector is designed for use in Zone 1 and Zone 2 or Zone 21 and Zone 22 hazardous areas, and is certified Ex db IIC T6 Gb and Ex tb IIIC T80°C Db for operation up to 70°C (158°F).

WARNING: POTENTIAL ELECTROSTATIC CHARGING HAZARD. The painted aluminium enclosure constitutes a potential electrostatic hazard and the equipment must only be cleaned using a damp cloth.

- The cable gland must be installed before use and must comply with the requirements of standards IEC 60079-0 and IEC 60079-1 with minimum IP66 ingress protection.
- The flameproof properties of the enclosure shall be maintained when an external alarm device is used.
- Unused cable entries must be sealed using an ATEX /IECEx Ex db and Ex tb certified stopping plug with minimum IP66 ingress protection.
- ATEX Certificate No.: TÜV 16 ATEX 7908 X
- IECEx Certificate No.: IECEx TUR 16.0035X
- INMETRO Certificate No.: UL-BR 26.0813X


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- Marking:
  Ex II 2 G Ex db IIC T6 Gb Tamb: -40°C≤Ta≤70 °C
  Ex II 2 D Ex tb IIIC T80°C Db
- This detector is designed for use in Zone 1 and Zone 2 or Zone 21 and Zone 22 hazardous areas, and is certified Ex db IIC T3 Gb -40°C ≤ Ta ≤ 70°C, Ex db IIC T4 Gb -40°C ≤ Ta ≤ 50°C and Ex tb IIIC T80°C Db for operation up to 70°C (158°F).

**WARNING:** POTENTIAL ELECTROSTATIC CHARGING HAZARD

- Only clean the equipment using a damp cloth.
- Operate the equipment only within certified hazardous areas
- Ensure all earthing connections are properly installed and maintained in accordance with the product manual.
- Use an insulating object when touching the equipment to prevent electrostatic discharge.
- Perform routine cleaning and maintenance outside hazardous areas when feasible.
- Do not touch the equipment when it is in an alarm or warning state.
- Shield the equipment from direct airflow in gas atmospheres and fast-moving particles in dust atmospheres, which may cause electrostatic charging on the enclosure surface.
- The equipment is not intended to be exposed to prolific electrostatic charge-generating mechanisms.
- Use only IP65 or IP66 rated cable glands to maintain ingress protection.
- When connecting an external alarm device, ensure the flameproof integrity of the enclosure is preserved.
- Seal all unused cable entries with ATEX/IECEx certified Ex db and Ex tb stopping plugs.
- The maximum service temperature at the cable entry point is 72.15°C, and at the branching point of the conductor is 73.75°C. The end user must select appropriate cables and cable glands rated for these temperatures and the intended application.
- ATEX Certificate No.: TÜV 16 ATEX 7908 X
- IECEx Certificate No.: IECEx TUR 16.0035X
- INMETRO Certificate No.: UL-BR 26.0813X
- Marking:
  Ex II 2 G Ex db IIC T3 Gb Tamb: -40°C ≤ Ta ≤ 70 °C
  Ex II 2 G Ex db IIC T4 Gb Tamb: -40°C ≤ Ta ≤ 50°C
  Ex II 2 D Ex tb IIIC T80°C Db

**UL CERTIFIED VERSION:**

- This detector is certified to Class I, Division 1, Groups B, C and D T6; Class II Division 1, Groups E, F and G T6 for operation up to 60°C (140°F)
- WARNING – POTENTIAL ELECTROSTATIC CHARGING HAZARD. The painted aluminium enclosure constitutes a potential electrostatic hazard and the equipment must only be cleaned using a damp cloth.
- The gas detector shall be installed in accordance with Article 501.10 (A) of the NEC, Class, I, Division 1, wiring method and Article 502.10 (A) of the NEC, Class, II, Division 1, wiring methods. To reduce the risk of ignition of hazardous atmospheres, a seal shall be installed within 18 in. of the enclosure.
- Unused cable entries must be sealed using UL Listed plugs having the same area classification Class I, Division 1, Groups B, C and D; Class II Division 1, Groups E, F and G under CCN EBNV.
- UL Certificate No.: UL-US-L147777-0002503217530-1
- Marking:

-40°C ≤ Ta ≤ 60°C

Class I, Division 1, Groups B, C, D T6

Class II Division 1, Groups E, F, G T6


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## 1.3 STORAGE INSTRUCTIONS

Some types of sensor available with Xgard Bright have limited life when left un-powered and/or may be adversely affected by temperature extremes or environmental contamination. Ideal storage conditions are 20˚C and 60% RH. Do not expose sensors to contaminants such as silicones, lead compounds and strong solvents such as isopropanol. It is strongly recommended detectors are installed and powered within 3 months of purchase.

## 1.4 CERTIFICATION LABEL

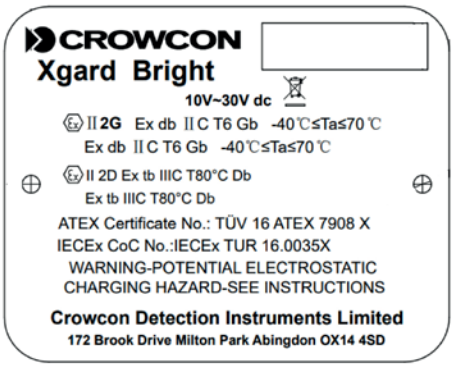



**Figure 1:** ATEX/IECEx Certification Label for Aluminium Enclosure

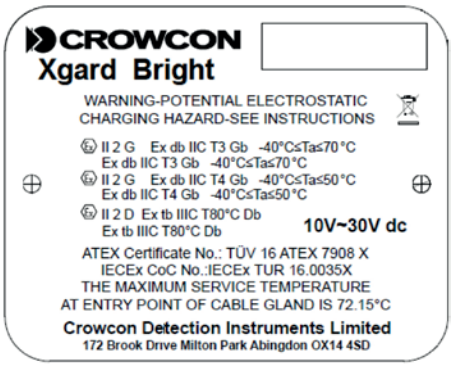



**Figure 2:** ATEX/IECEx Certification Label for Stainless Steel Enclosure




**Figure 3:** UL Certification Label




**Figure 4:** Warning Label

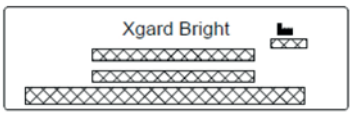
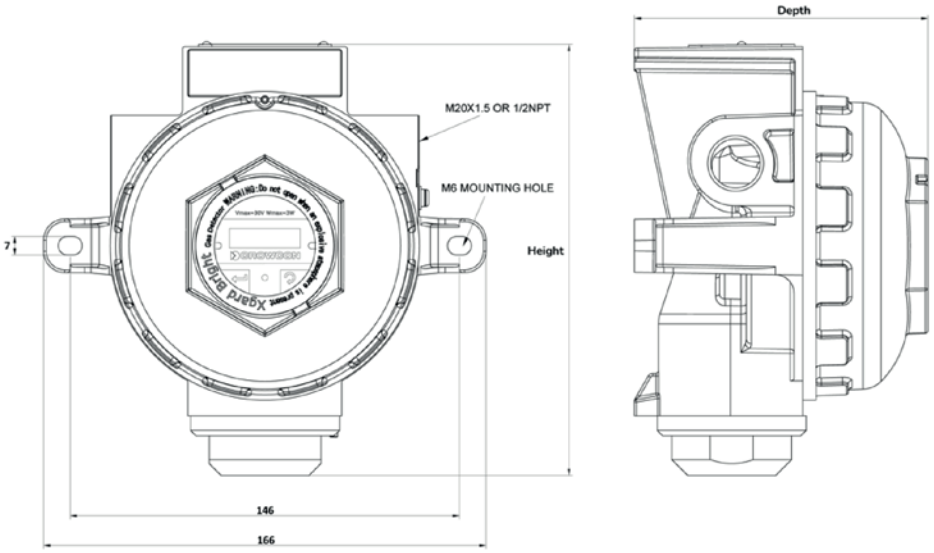



**Figure 5:** Gas Label


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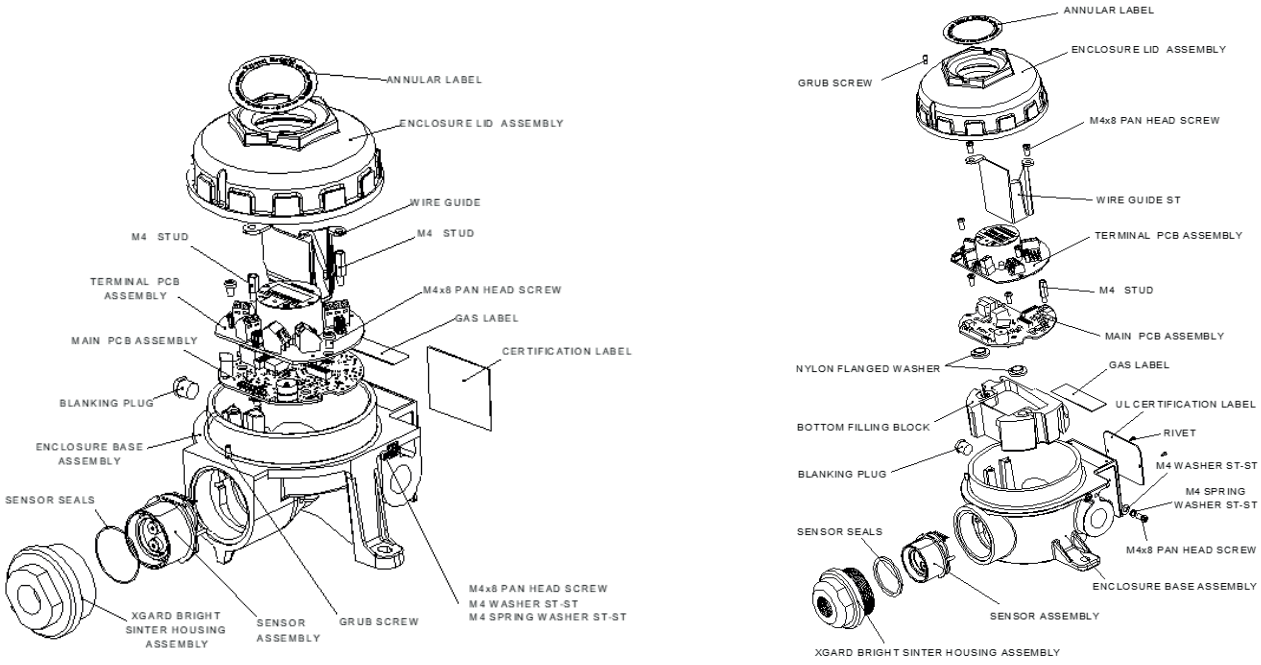
## 1.5 PRODUCT DIMENSIONS




| Height: | |
|---|---|
| ATEX/IECEx AL | 156 |
| ATEX/IECEx SS | 165 |
| UL AL | 162 |
| UL SS | 162 |
| **Depth:** | |
| ATEX/IECEx AL | 109 |
| ATEX/IECEx SS | 105 |
| UL AL & SS | 110 |
| UL AL & SS | 103 |

**Figure 6:** Xgard Bright dimensions (mm)

## 1.6 EXPLODED VIEW




**All Versions Except UL Stainless Steel** **UL Stainless Steel**

**Figure 7:** Xgard Bright exploded view


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# 2. INSTALLATION

WARNING: Installation must be in accordance with the recognized standards of the appropriate authority in the country concerned. For further information please contact Crowcon. Prior to carrying out any installation work ensure local regulations and site procedures are followed.

## 2.1 LOCATION

The detector should be mounted where the gas to be detected is most likely to be present. The following points should be noted when locating gas detectors:

- To detect gases which are lighter than air, detectors should be mounted at high level and Crowcon recommend the use of a collector cone.
- To detect heavier than air gases, detectors should be mounted at low level.
- When locating detectors consider the possible damage caused by natural events e.g. rain or flooding. For detectors mounted outdoors Crowcon recommend the use of a Spray Deflector.
- Consider ease of access for functional testing and servicing.
- Consider how the escaping gas may behave due to natural or forced air currents. Mount detectors in ventilation ducts if appropriate.
- Consider the process conditions. For example, butane is normally heavier than air, but if released from a process which is at an elevated temperature and/or pressure, the gas may rise rather than fall.
- Location of oxygen sensors requires knowledge of the gas that may displace the oxygen. For example, carbon dioxide is denser than air and therefore is likely to displace oxygen from low levels upwards.
- Sensors should be mounted at head height (1.5m nominally) to detect gases of a similar density to air, assuming that ambient conditions and the temperature of the target gas are nominally 20˚C.

The placement of sensors should be determined following advice of experts having specialist knowledge of gas dispersion, the plant processing equipment as well as safety and engineering issues. The agreement reached on the locations of sensors should be recorded.

## 2.2 MOUNTING

Xgard Bright should be installed at the designated location with the sensor pointing down. This ensures that dust or water will not collect on the sensor and stop gas entering the cell. Care should be taken when installing the detector to avoid damaging the painted surface of the enclosure.
There are two M20x1.5 (or ½" NPT) entry ports on the base. One entry port will be used for power supply input during normal operation. Unused port will be blocked by blind plug, or can be used to connect external alarm device or be used for connecting devices to the multi drop communications. End user will only use certified cable gland for installation.

Refer to the dimension on Figure 6 for mounting hole details.

## 2.3 INTERNAL ELECTRICAL CONNECTIONS

All electrical connections are made via the screw terminal block mounted on the terminal PCB in the enclosure base. The terminals are marked '+', 'Sig' and '-' and correct polarity should be observed when connecting the detector to control equipment.

Xgard Bright is factory set as a 'current sink' device unless otherwise specified when ordering. To reset to 'current source', open the enclosure base and move the three links on the terminal PCB from the 'SINK' position to the 'SOURCE' position.

Relays are fitted to indicate any Fault, Level 1 alarm or Level 2 alarm and are marked accordingly. The Alarm relays are normally de-energised, the Fault relay is normally energised (fail-safe). The volt-free contacts can be changed from Normally Open (NO) to Normally Closed (NC) by moving the adjacent link, where NO and NC refer to the de-energised state. The fault relay on Xgard Bright is normally energised (when the detector is healthy), therefore the operation of the FAULT contacts is the reverse of that stated.

To limit radio frequency interference, the enclosure base and cable armour should be grounded (earthed) at the control panel. Ensure the earth connection is provided in a safe area only, so as to avoid earth loops.


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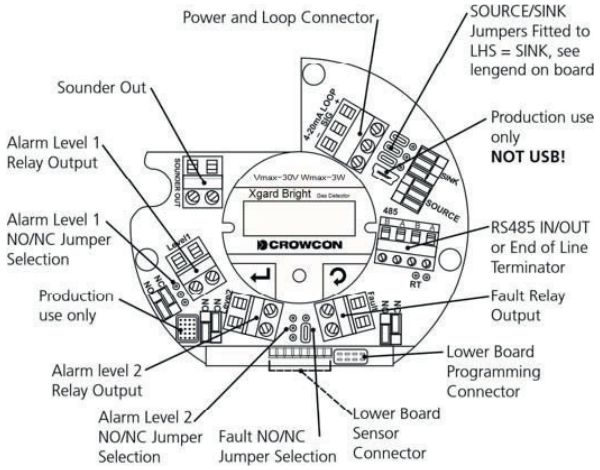



**Figure 9:** Internal electrical connections

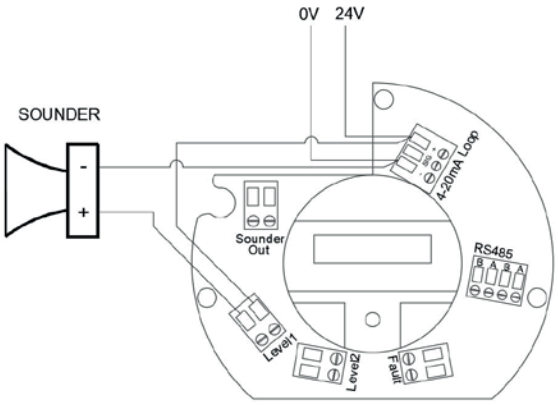



**Figure 8:** Sounder/Relay Wiring

Sounder terminals provide an open-collector output for directly triggering audible-visual alarms. Alarms will be activated by a voltage equivalent to the Xgard Bright detectors supply (24Vdc nominally, minimum 12 Vdc). Devices with a total maximum current of 250mA may be connected. RS-485 terminals can be used to enable Xgard Bright to communicate to a Modbus Master device using the Modbus RTU protocol. This function can be used in conjunction with the 4-20mA analogue signal to transmit data to a central system, or used for multi-dropping detectors on an addressable network. An 'Xgard Bright Modbus Instructions' document is available from Crowcon detailing all necessary connection, data format and address information.

WARNING: The mini USB socket is not intended for customer use, connecting this to a computer is likely to damage both Xgard Bright and the computer.

Sounders connected to Sounder Out port will not work if the supply voltage drops below 12Vdc. An alternative connection method for directly connecting local sounder is to connect it via the Alarm 1 relay as shown in Figure 14.

If the detector is connected to a control system the 4-20mA 'Signal' wire should be connected as normal. If the detector is installed for use in 'stand-alone' mode (i.e. not connected to a control system), the SINK/SOURCE links must be set as shown in Standalone Operation section.

Xgard Bright is available as an option with HART Protocol Revision 7 communications for integration with Asset management Systems or checking with a HART-enabled hand-held communicator. The user can access HART common and universal commands supported by HART 7, no DD (Device Description) files are required for these commands.

HART enabled Xgard Bright detectors can also be multi-dropped on a common cable in a 'daisy chain' configuration to a suitable controller. A unique address must be set on each detector using a HART host (ie 1,2,3 etc). Detectors with a HART address set as integers other than zero will maintain their signal current output at 4mA (loop disabled).

## 2.4 GENERAL CABLING REQUIREMENT

Cabling to Xgard Bright must be in accordance with the recognised standards of the appropriate authority in the country concerned and meet the electrical requirements of the detector.

Suitable explosion proof glands must be used. Alternative cabling techniques, such as steel conduit, may be acceptable provided appropriate standards are met.

Xgard Bright requires a dc supply of Class 2 10-30V, at up to 100mA. Ensure there is a minimum of 10 V at the detector, taking into account the voltage drop due to cable resistance. For example, a nominal dc supply at the control panel of 24V has a guaranteed minimum supply of 18V. The maximum voltage drop is therefore 8 V. Xgard Bright can demand up to 100mA and so the maximum loop resistance allowed is 80Ohms.

A 1.5mm2 cable will typically allow cable runs up to 3.3km. Table 1 below shows the maximum cable distances given typical cable parameters for this example calculation.


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| C.S.A. | | RESISTANCE (OHMS PER KM) | | MAX. DISTANCE (KM) |
|---|---|---|---|---|
| mm2 | Awg | Cable | Loop | |
| 1.0 | 17 | 18.1 | 36.2 | 2.2 |
| 1.5 | 15 | 12.1 | 24.2 | 3.3 |
| 2.5 | 13 | 7.4 | 14.8 | 5.4 |

**Table 1:** Maximum cable distance for typical 3 core cables

The acceptable cross sectional area of wire used is 0.5 to 2.5mm2 (20 to 13awg). The maximum acceptable diameter of cable used is 15mm. The table is provided for guidance only, actual cable parameters for each application should be used to calculate maximum cable distances.

### FOR 4-20 MA CURRENT LOOP

Fulfils the requirements for 4 to 20 mA current loop and HART connections, allows for connection and powering of accessory beacon or sounder subject to current consumption, cable resistance and panel voltage. Current consumption should consider worst case e.g. when the accessories are powered.

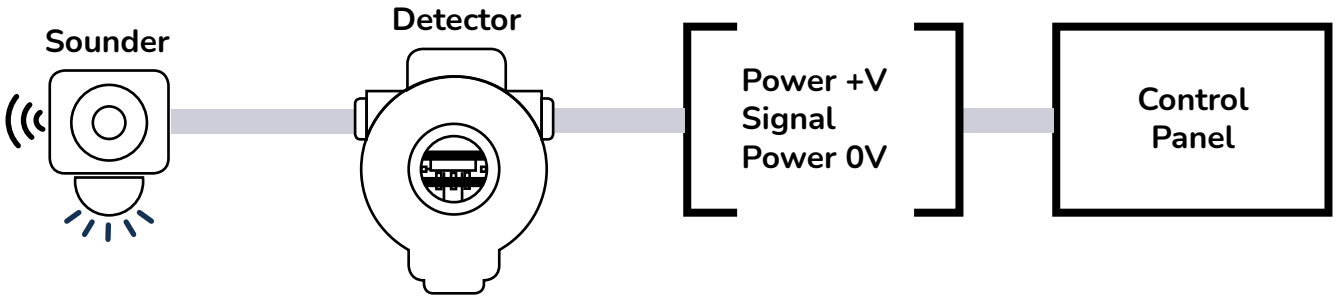



**Figure 10:** Typical 4-20 mA connection

### EXAMPLE CALCULATION 1

What is the longest cable for a bright to operate using point to point connection and powering a sounder with 250mA current consumption. Use parameters of 1.5mm2 cable, where the controller has a guaranteed minimum output voltage of 18V.

This type of cable has resistance of 12.1Ω/km, therefore the there and back cable resistance is 24.2. Xgard Bright has min voltage requirement of 10V.

The alarm 2 current for Xgard Bright (Catalytic) is 95mA, Xgard Bright (IR) is 90mA, Xgard Bright (MEMS) is 34mA and the sounder output max current is 0.25 A, so a total current for the Catalytic in alarm driving the sounder output is:

Max current = 0.25 + 0.095 = 0.345 A.

18V = 10V + (0.345 x 24.2 x d), where d is distance in km

d = (18 – 10) / (0.345 x 24.2) = 0.958 km

### EXAMPLE CALCULATION 2

Cabling to Xgard Bright must be in accordance with the recognised standards of the appropriate authority in the country concerned and meet the electrical requirements of the detector.

Suitable explosion proof glands must be used. Alternative cabling techniques, such as steel conduit, may be acceptable provided appropriate standards are met.

As example calculation 1 but without the sounder.

Xgard Bright Catalytic requires a dc supply of 10-30V, at max current in alarm 2 of 95mA. Ensure there is a minimum of 10V at the detector, taking into account the voltage drop due to cable resistance. For example, a nominal dc supply at the control panel of 24V has a guaranteed minimum supply of 18V. The maximum voltage drop is therefore 8V. Xgard Bright can demand up to 95mA and so the maximum loop resistance allowed is approx. 80Ω.


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**FOR MULTIDROP MODBUS**

This fulfils the requirements of multidrop communications back to a compatible addressable control panel. Due to current consumption of multiple detectors powering of accessories via the detector sounder/beacon output or relay contact must be avoided.

Each detector must be configured with a unique node address when connected in an addressable network.

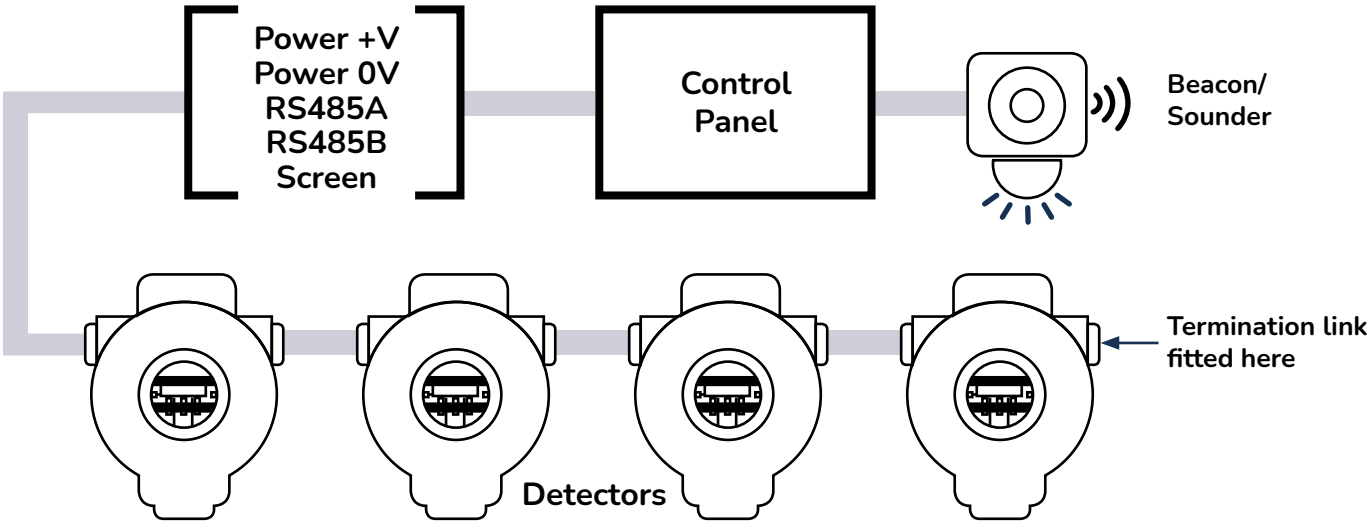



**Figure 11:** Multi-drop Modbus connection

Four connections are required for multidrop operation: a 24V/0V dc power supply, and RS-485 A and B connections to the appropriate terminals. Two sets of RS-485 terminals and a spare cable gland entry (sealed with a stopping gland by Crowcon) are provided to enable signals to be 'looped' to the next detector easily.

To minimise cable voltage drops (and to maximise the potential total cable length and detector network quantity) large cross-sectional area (c.s.a.) cable must be used for the 24V/0V power connection. Crowcon recommends cable with 1.5mm2 conductors is used for the power.

Twisted pair and screened cable is recommended for the RS485 signals. The screen is to be earthed at the control panel only, but continuity must be maintained through the detectors extending to the end of line detector. The end of line detector also needs a terminating resistor link fitted to the top PCB (the terminals labelled RT).

Specialist cables are available combining large c.s.a. conductors for power and twisted-pair signal cables for RS-485 communications, however in some cases it may be necessary to run separate cables to the detector network. In this instance it may be most practical to terminate the two cables within a junction box near to each detector, and drop and single/combined cable with smaller power conductors locally to the detector.

On large networks, or where long cable runs are required, it may be necessary to power groups of detectors via separate power supplies placed locally around the installation.  Where this method is deployed, the 24V/0V cables for each group of detector must be isolated to their dedicated local power supply.

**CALCULATING ACCEPTABLE CABLE LENGTH AND DETECTOR QUANTITIES**

It is essential before attempting installation to calculate the voltage to each detector given the power supply voltage, cable resistance and cable lengths required. The more detectors connected to the linear bus, the greater the power required to run the system. To calculate the power required for a particular setup, it is necessary to know the cable resistance between each pair of detectors. A current of a maximum 0.07A (toxic) must be allowed for each 'hop' between each detector (this assumes the highest power configuration for each detector: Catalytic sensor). The voltage to be applied can be calculated by estimating the voltage drop across each detector 'hop' – at the end at least 10V must remain to ensure that the last Xgard Bright detector functions correctly.

Follow the steps outlined below and the sample calculation shown in the next section to calculate for specific applications:

1. The voltage must not fall below 10V, so start the calculation by setting the voltage at the last detector in the line at that value.
2. Each detector may draw up to 0.070A. Calculate the cable voltage loss of the first 'hop' between detectors by taking the current 0.070A and multiply this by the cable resistance of the 'hop' between the last and the last but one detector.


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3. Add this voltage drop to the initial 10V to get the lowest acceptable voltage at the last but one detector. Add 0.070A to the value for the 'aggregate current' to get to 0.14A, the minimum current running through the last but one 'hop' of the bus. Multiply this by the cable resistance for the last but one 'hop' to get the next voltage drop.
4. Repeat this process for each detector, accumulating the voltage losses that will occur between each detector.
5. The maximum detector voltage of 30V must not be exceeded.

### EXAMPLE CALCULATION USING THE ABOVE RULES

How many detectors can be put on a multidrop cable if:

1. The controller has a guaranteed minimum output voltage of 18V.
2. Cable resistance is 12.1Ω/km.
3. There is 20m between each detector and 20 m from the final detector to the controller.
4. The worst case current draw (Xgard Bright toxic) is 70mA.

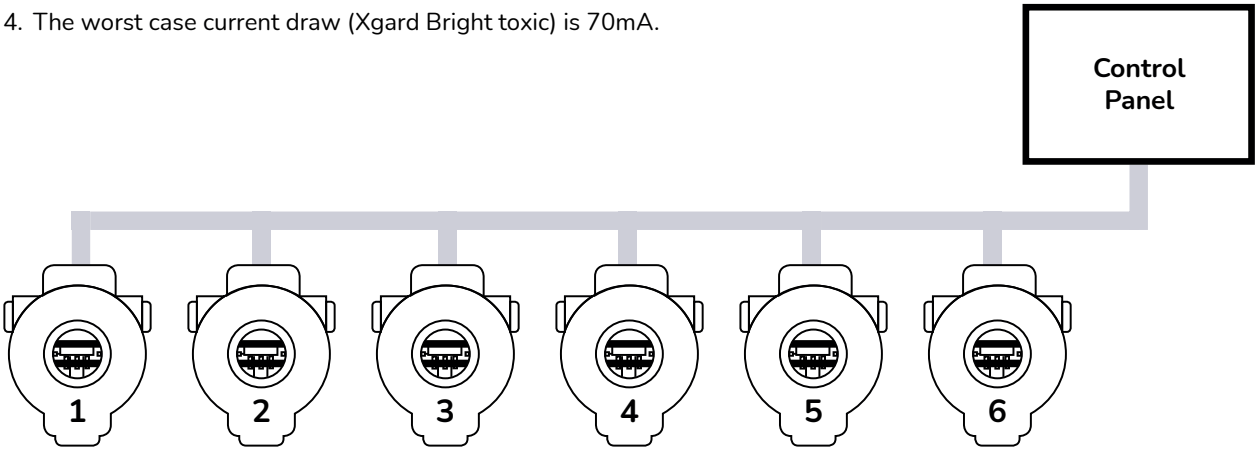



**Figure 12:** Xgard Bright on a Modbus loop

So consider the voltage to the detector furthest (n=1) from the controller has to be 10V. Each cable segment has a there and back resistance of 12.1 x 2 x 20/1000 = 0.484 ohms.

So the cable volts drop to detector (n=2) is:
Vc = 0.070 x 0.484 = 0.03388V

V(n=2) = V(n=1) + Vc = 10.0338 V

Now the voltage at detector (n=3) is

V(n=3) = V(n=2) + 2Vc (as there is twice the current supplied through this cable segment)

V(n=3) = 10.03388 + 0.06776 = 10.10164 V

Tabulating the results for each detector position we get:

| DETECTOR | RESISTANCE (OHMS PER KM) | CABLE CURRENT (A) | CABLE VOLTAGE DROP (V) |
|---|---|---|---|
| N=1 | 10 | 0.070 | 0.03388 |
| N=2 | 10.03388 | 0.14 | 0.06776 |
| N=3 | 10.10164 | 0.21 | 0.10164 |
| N=4 | 10.20328 | 0.28 | 0.13552 |
| N=5 | 10.3388 | 0.35 | 0.1694 |
| N=6 | 10.5082 | 0.42 | 0.20328 |
| N=7 | 10.71148 | 0.49 | 0.23716 |
| N=8 | 10.94864 | 0.56 | 0.27104 |
| N=9 | 11.21968 | 0.63 | 0.30492 |
| N=10 | 11.5246 | 0.7 | 0.3388 |


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| DETECTOR | RESISTANCE (OHMS PER KM) | CABLE CURRENT (A) | CABLE VOLTAGE DROP (V) |
|---|---|---|---|
| N=11 | 11.8634 | 0.77 | 0.37268 |
| N=12 | 12.23608 | 0.84 | 0.40656 |
| N=13 | 12.64264 | 0.91 | 0.44044 |
| N=14 | 13.08308 | 0.98 | 0.47432 |
| N=15 | 13.5574 | 1.05 | 0.5082 |
| N=16 | 14.0656 | 1.12 | 0.54208 |
| N=17 | 14.60768 | 1.19 | 0.57596 |
| N=18 | 15.18364 | 1.26 | 0.60984 |
| N=19 | 15.79348 | 1.33 | 0.64372 |
| N=20 | 16.4372 | 1.4 | 0.6776 |
| N=21 | 17.1148 | 1.47 | 0.71148 |
| N=22 | 17.82628 | 1.54 | 0.74536 |
| **Power Supply** | **18.57164** | | |

**Table 2:** Calculating cable length & detector quantities

So 22 detectors just exceeds the power supply guarantee voltage, therefore the answer for a safe maximum number of detectors is 21. If this is not a convenient solution then there is scope for increasing the number by changing the power supply or using thicker (lower resistance) cable.

## 2.5 EARTHING REQUIREMENTS

Earth terminals are provided on the outside of the Xgard Bright enclosure adjacent to the top-right cable entry, and internally adjacent to the left-hand sounder out cable connector. For electrical safety it is essential that the Xgard Bright enclosure is bonded to earth, usually using the external earth lug, if an earth cable is provided in the field cable the internal earth point can be used. The grounding terminal connection must be tightened using a torque screwdriver to 10Nm; secured using an M4 x 6mm screw, a plain washer and a star/lock washer. Earthing cables must be 4mm2 (AWG 10 in the USA) or greater in cross-sectional area.

For use in North America, the internal grounding terminal shall be used as the equipment grounding means and that the external terminal is only a supplemental bonding connection where local authorities permit or require such a connection.

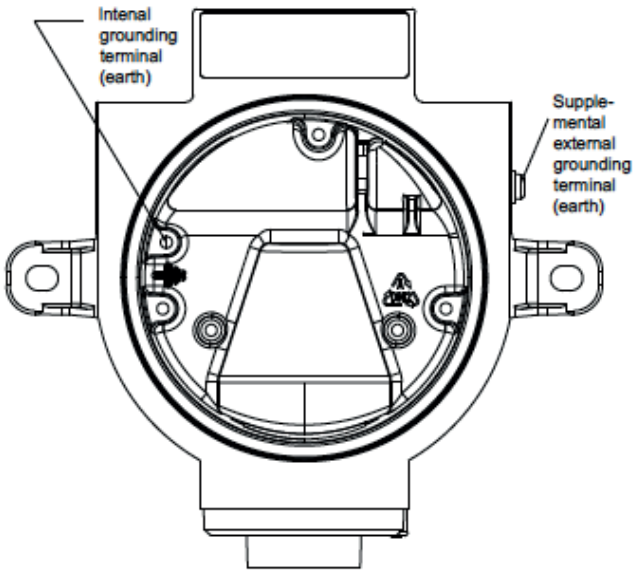



**Figure 14:** Earthing Connection on Aluminium Enclosure

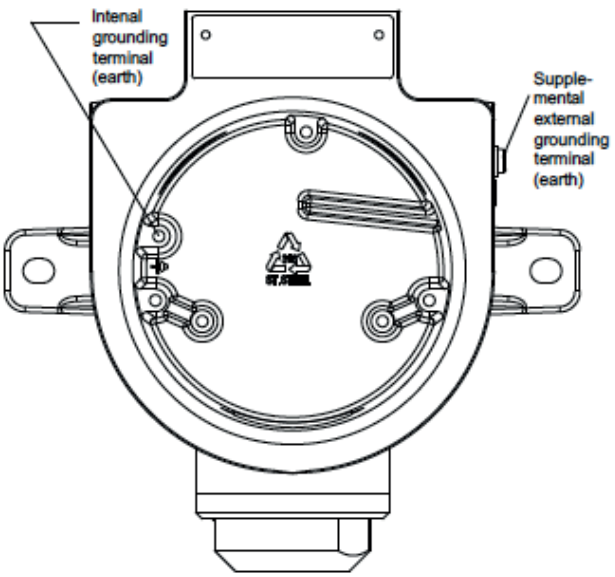


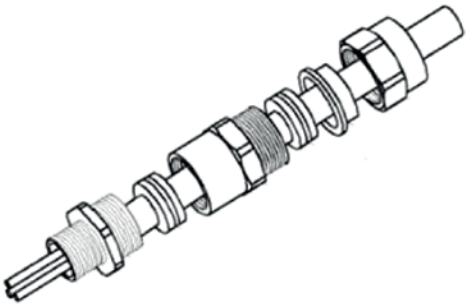

**Figure 13:** Earthing Connection on Stainless Steel Enclosure


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## 2.6 CABLE GLANDS

Instructions for installing appropriate certified glands with unarmoured and armoured cable is given below.

**UNARMOURED CABLE**



| | |
|---|---|
| 1 | Backnut |
| 2 | Compression Spigot |
| 3 | Seal |
| 4 | Middle Nut |
| 5 | Inner Seal |
| 6 | Entry |

**Figure 15:** Unarmoured cable

Allow sufficient length of cable to the detector, fit shroud if required, pass the cable through cable gland.

1. Screw the gland entry into the Xgard Bright cable Entry. Using two spanners or wrenches, hold the gland Entry in position to prevent rotation and tighten the Middle Nut until resistance is felt between the Seal and cable. Then turn the Middle Nut through a further half to one full turn to complete the Inner Seal.

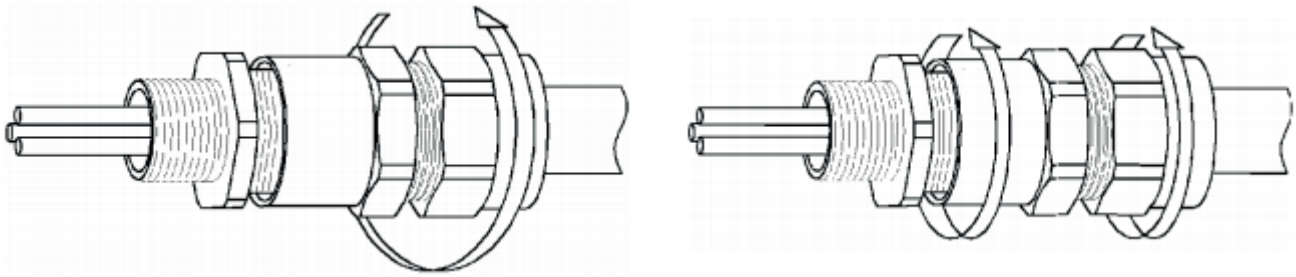
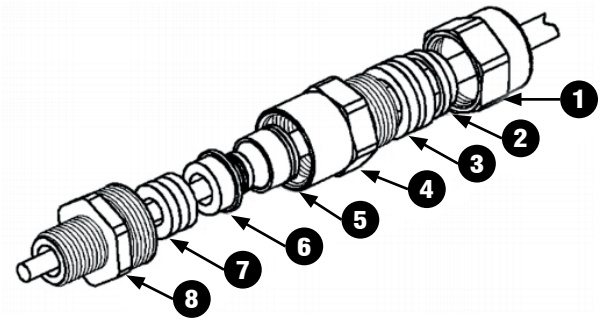


**Figure 16:** Cable Gland Installation Instructions

2. Hold the Middle Nut in position to prevent rotation and tighten the Backnut until resistance is felt between the Seal and cable, then turn the Middle Nut through a further half to one full turn to complete the Outer Seal.

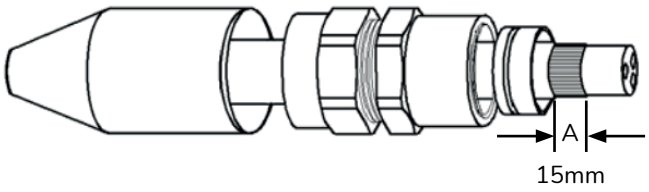
**ARMOURED CABLE**



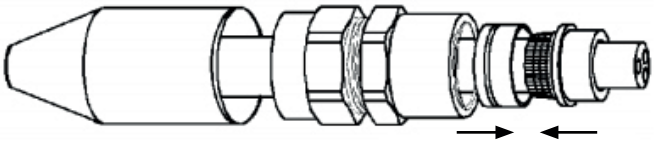

| | |
|---|---|
| 1. | Backnut |
| 2. | Compression Spigot |
| 3. | Compression seal |
| 4. | Middle Nut |
| 5. | Reversible Armour Clamping Ring |
| 6. | Armour Spigot |
| 7. | Inner seal |
| 8. | Entry |


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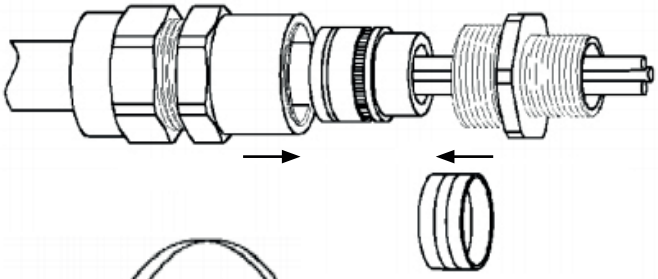
1. Expose the armour/braid “A” by stripping the cable's armoured / woven layer and removing the cable filler.



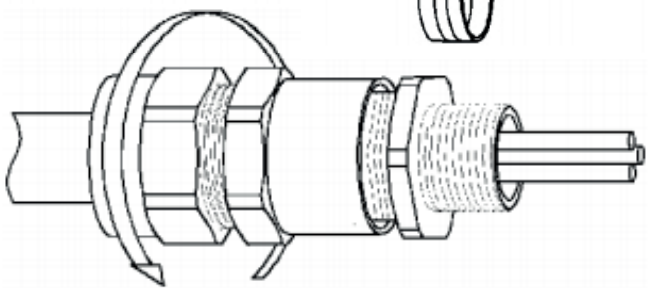

2. Push the cable through the Armour Spigot. Spread armour/braid over the Armour Spigot until the end of the armour/braid is up against the shoulder of the armour cone. Position the Armour Clamping Ring.



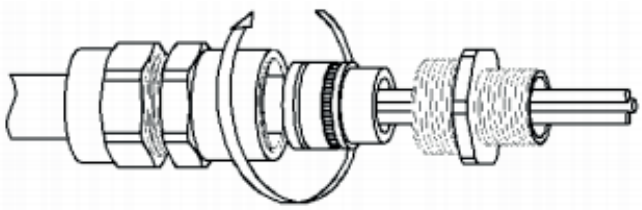
3. Remove the Inner seal from the Entry. Place the Entry over the Armour Spigot. Move the Middle Nut to meet the Entry.



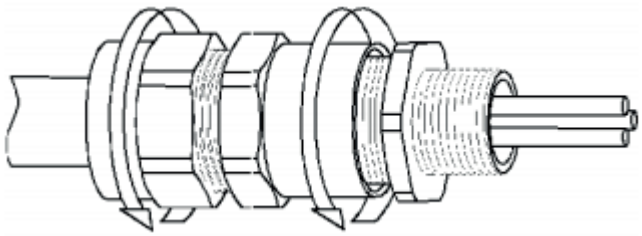
4. Hold the Entry in position with a spanner/wrench to prevent rotation. Hand tighten the middle Nut to the Entry and turn a further half to one full turn with a spanner/wrench.



5. Unscrew the Middle Nut and visually inspect that the armour/braid has been successfully clamped between the Armour Spigot and the Armour Clamping Ring. If armour/braid not clamped, repeat assembly.



6. Remove Entry and refit Inner Seal, replace Entry and re-assemble Middle Nut onto the entry component. Tighten up the Middle Nut by hand then using a wrench/spanner a further 1 to 4 turns until fully tight.



NOTE:

1. These instructions are provided for general guidance only. It is essential that the instructions provided by the cable gland manufacturer are strictly adhered to.
2. Only cable glands ATEX and IECEx certified Exd Flameproof may be used with Xgard Bright.
3. Cable glands must have a minimum ingress protection rating of IP66.


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# 3. OPERATION

WARNING: Prior to carrying out any work ensure local regulations and site procedures are followed. Never attempt to open the detector or enclosure base when flammable gas is present. Ensure that the associated control panel is inhibited so as to prevent false alarms.

## 3.1 OPERATION PANEL

The Xgard Bright operation panel comprises an OLED screen, a three-colour status LED and two magnetically operated Hall Effect switches. The screen displays white characters on a black background and can be viewed clearly even in bright sunlight. Reversed white screen saver will be activated in normal detection condition while long time no operation.

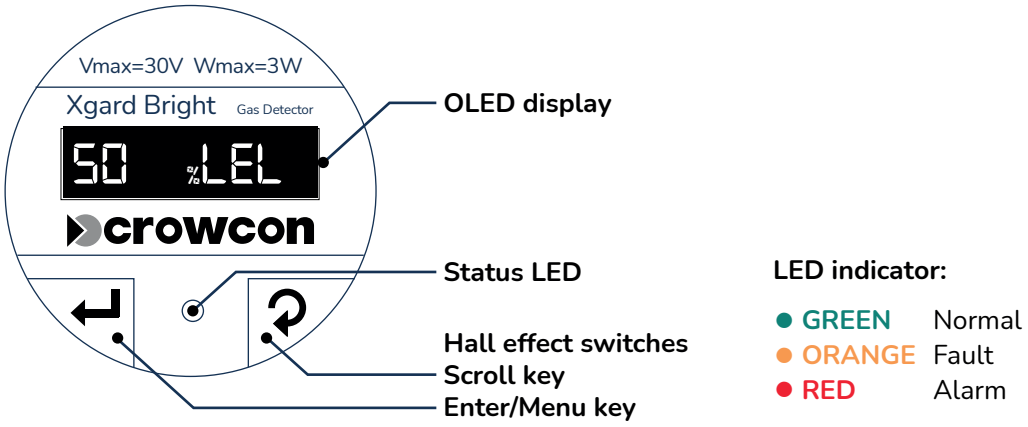



**Figure 18:** Xgard Bright operation panel

## 3.2 KEY OPERATION

Key response has time dependency, two kinds of action would be generated depend on how long time the key was held.

- Short-time action event, magnet applied and removed in 2 seconds.
- Long-time action event, magnet applied and hold for more than 2 seconds, a short beep will indicate 2 seconds is complete.

## 3.3 START UP




**Figure 19:** Start up & warm up

When the Xgard Bright is powered up, the unit will perform internal diagnostic checks whilst the display will show a Crowcon logo. This procedure will be displayed for about 10 seconds followed by displaying a warming up status for about 120 seconds. If the diagnostic checks were successful, the gas status screen will be displayed. In normal operation the gas level will be indicated on the display. Use the supplied magnetic wand to select the Scroll key to move between the information screens shown over the page.


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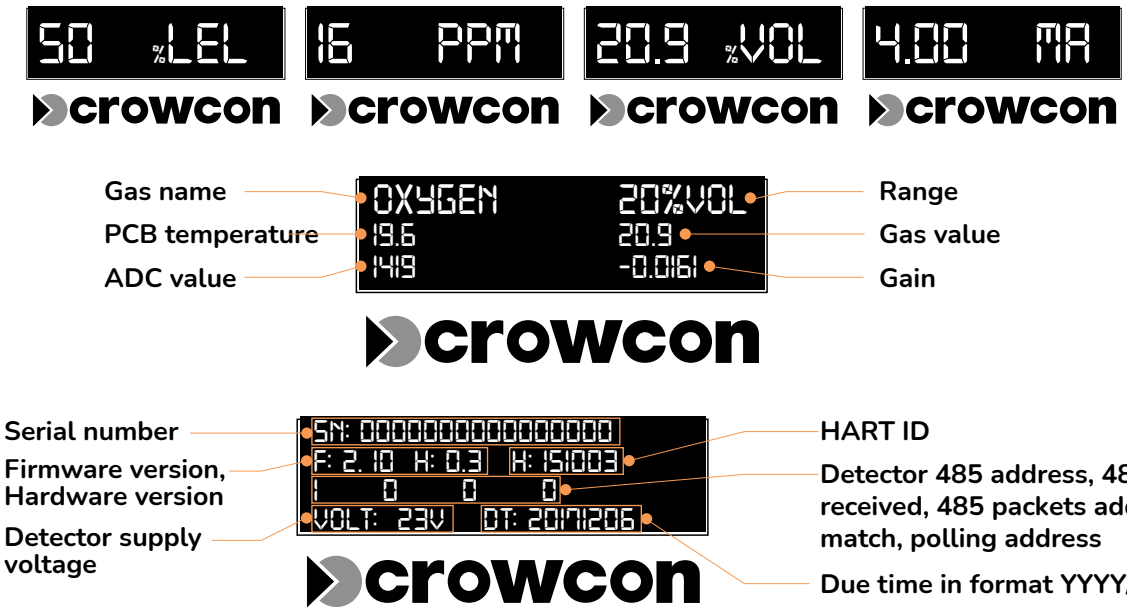



**Figure 20:** Gas status screen

For optimal performance, all sensors generally require a stabilization time of 10-30 minutes. For biased sensors, such as the long-life O2 sensor, a longer stabilization period of approximately 3 hours is necessary before calibration. In the event of power losses:

- 120 seconds or less: Allow a 2-hour stabilization period.
- Exceeding 120 seconds: A 3-hour stabilization period is required.

Be aware that false alarms may occur within the first hour after power-up or re-initialization. To maintain accurate readings, ensure continuous power supply to the sensors. It is recommended to use a UPS system to prevent power interruptions.

## 3.4. MENU FUNCTIONS

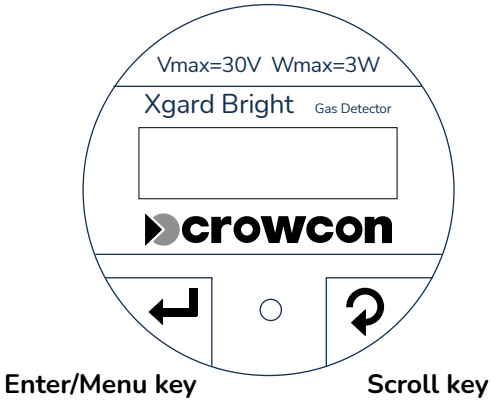



**Figure 21:** Xgard Bright Interface

From the gas display screen, use the supplied magnetic wand to select the Enter/Menu key to show password screen.

Enter the password: '0000' to display the following menu. Select the Scroll key to change the displayed number, and the Enter/Menu key to move to the next integer. Hold the Enter/Menu key for two seconds to enter the menu.


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Use Scroll key to move down the menu options list and to change the numbers in any screen. Hold the scroll key for 2 seconds to move up the menu options.

1. Zero: to zero the sensor
2. Cal gas: to calibrate the sensor
3. To main menu: returns the display to the main gas screen
4. Set Alarm 1: to adjust the first alarm threshold
5. Set Alarm 2: to adjust the second alarm threshold
   NOTE: if the Alarm 2 threshold is set lower than the Alarm 1 threshold both will act as falling alarms.
6. Chn/Eng: switch between Chinese and English display
7. Set 485 Addr: sets the node address of the detector for Modbus operation
8. Test Relay: adjust the analogue output signal: Dec/OK to decrease, Add/ Back to increase. The alarm relays will activate at the appropriate point.

a. Adj Loop mA: errors in the zero-level analogue output signal can be adjusted by up to +/-2mA.

Hold Enter/Menu key for 2 seconds to enter menu or confirm changes to (eg) alarm levels. After applying the magnet a short beep will indicate when 2 seconds is complete and the magnet can be removed. A long beep will then confirm the change.

Hold the Enter/Menu key for 2 seconds to exit the menu from any point.

## 3.5 ZERO

NOTE: Purge sensor with the appropriate gas before zeroing.

This function should only be carried out in 'clean air' and allows the Xgard Bright to be zeroed at any time. To carry out the zero function, press 'Next'. The zeroing screen will be displayed. When zeroing is complete a pass or fail screen will be displayed. Press 'OK' to return to the Main menu.




**Figure 22:** Zero Calibration Start Screen

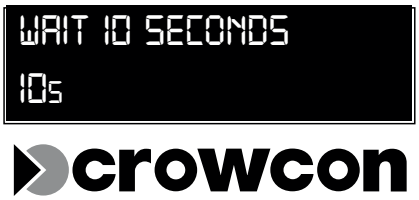



**Figure 23:** Zero Calibration Wait Screen

NOTE: Calibration of oxygen channels: Zeroing is not required for an oxygen instrument and selecting zero will not function.

## 3.6 CALIBRATE

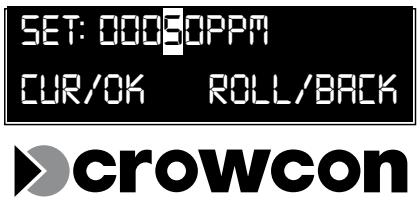



**Figure 24:** Calibration level screen

In main menu status chose 'Cal gas', the calibration level screen will be displayed. To cancel the calibration, long press the right key '/Back', The Main menu will be displayed. Adjust the value shown to match the calibration gas concentration, to continue, long press left key '/OK' to accept the calibration level.


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Connect the calibration cap to the Xgard Bright and then connect the appropriate gas cylinder to the adapter. When the gas reading is stable press the right key 'Mark'. To complete calibration press left key 'DONE'. If the calibration was successful the success symbol will be displayed and return to the main menu.




**Figure 25:** Calibration completion screen

Please refer to the table below to set span value while testing or calibrating flammable IR sensors. Please refer to technical note XGB015 regarding Xgard Bright IR sensor calibration for more information.

| SENSOR | TEST GAS CONCENTRATION | SPAN VALUE |
|---|---|---|
| IR Methane (ATEX/IECEx) | 2.2% Vol. Methane | 50% LEL |
| IR Methane (UL) | 2.2% Vol. Methane | 44% LEL |
| IR Propane (ATEX/IECEx) | 0.85% Vol. Propane | 50% LEL |
| IR Propane (UL) | 0.85% Vol. Propane | 40.5% LEL |
| IR Butane (ATEX/IECEx or UL) | 0.85% Vol. Propane | 40.5% LEL |
| IR Pentane (ATEX/IECEx or UL) | 0.85% Vol. Propane | 40.5% LEL |
| IR LPG (ATEX/IECEx or UL) | 0.85% Vol. Propane | 40.5% LEL |

**Table 3:** IR Sensor Calibration Values

For MEMS flammable sensor, zero calibration must be performed in clean background air (do not use zero air or nitrogen gas). To test (or calibrate if required), ensure the test gas mixture is 2.2% Vol. Methane (CH4), 20.44% Vol. Oxygen (O2), 0.91% Vol. Argon and Nitrogen (N2) balance. Please refer to technical note GEN110 regarding MEMS sensor calibration for more information.

## 3.7 STANDALONE OPERATION

Xgard Bright will go into a 'F' (fault) state if the processor does not measure a valid analogue output signal (for example, if the detector is not connected to a control system and the detector relays are just being used for activating a local sounder/beacon).

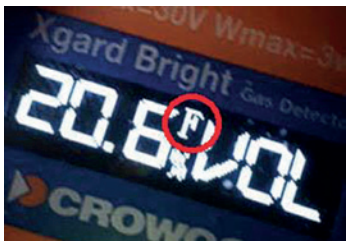



**Figure 26:** Fault Indication on Screen

If the detector is to be used in stand-alone mode as described above, a loopback of the analogue signal must be configured for the fault state to be cleared. This can be achieved in two ways as described as external and internal loopback below.

### EXTERNAL LOOPBACK

To create an external loopback, a 6k8Ω load resistor can be fitted to enable a 4mA signal to be measured at the 'SIG' terminal by the processor.


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If the detector is set to Current 'Sink', the resistor should be fitted between the 'SIG' and '+' terminals. And if the links are set to Current 'Source' the resistor should be fitted between the 'SIG' and '–' terminals.

**INTERNAL LOOPBACK**

If the detector is to be used in 'stand-alone' mode as described above, set the SINK/SOURCE jumper links to create an internal loopback of the analogue signal, as show in the figure.

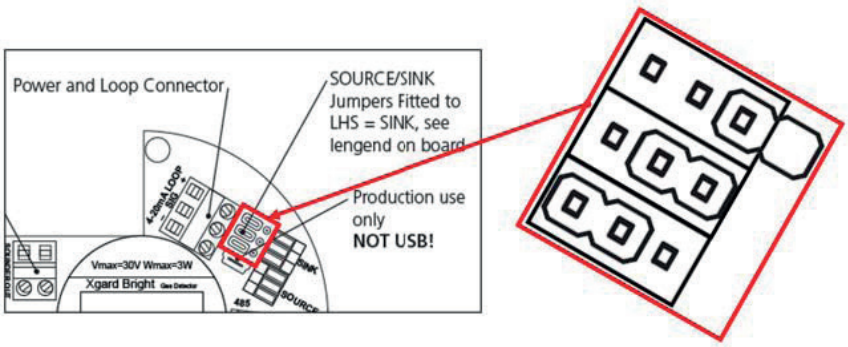



**Figure 27:** Link settings for standalone mode

WARNING: The analogue output (4-20mA) is disabled when internal loopback is configured, the output will remain 0mA. Use the external loopback if analogue output is required or reconfigure the detector for a SINK/SOURCE output.

## 3.8. ROUTINE MAINTENANCE

WARNING: prior to carrying out any work ensure local regulations and site procedures are followed. Never attempt to unscrew the lid or sensor retainer with power applied or when flammable gas may be present. Ensure the associated control panel is inhibited so as to prevent false alarms.

Service and maintenance work should be carried out by Crowcon, an approved service centre or by suitably qualified and trained personnel. Site practices and conditions will dictate the frequency with which detectors are tested and calibrated. Crowcon recommends that detectors are gas tested at least every 6 months (12 months for MPS and IR sensors) and re-calibrated as necessary. Do not use damaged components or damaged electrochemical cells.

The sinter should be inspected regularly, and replaced if it has become contaminated. A blocked sinter may prevent gas from reaching the sensor. Ensure that the sensor retainer and junction box lid O-rings are present and in good condition in order to maintain the ingress protection of the product.

## 3.9. SENSOR REPLACEMENT

The operational life of sensors depends on the application and amount of gas to which the sensor has been exposed.

| SENSOR | OPERATING LIFE EXPECTANCY |
|---|---|
| Oxygen (standard) | 2 years |
| Oxygen (long-life) | 5 years |
| Electrochemical (toxic) | 2-3 years |
| Catalytic (flammable) | 3-5 years |
| MEMS (flammable) | 10+ years |
| IR | 5+ years |

**Table 4:** Sensor Operating Life Expectancy under Normal operating Conditions


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Catalytic sensors can suffer from loss of sensitivity when there is a presence of poisons or inhibitors such as silicones, sulphides, chlorine, lead or halogenated hydrocarbons. Crowcon use poison resistant catalytic sensor to maximise operational life, however in applications where such compounds are present we recommend the use of MEMS sensors which are immune to such poisons and inhibitors.

Toxic and flammable gas sensor modules should be replaced when they fail to calibrate or show a very low response to test gas. Oxygen sensors must be replaced every two years.

To replace a sensor module:

1. Switch off and isolate power to the detector.
2. Loosen the grub-screw and unscrew the sensor retainer.
3. Carefully withdraw the sensor module by grasping the top and bottom of the moulding.
4. Fit the replacement sensor (having checked that the part number matches that stated on the detector junction box label). Take care to align the locating pins correctly with the slots in the Xgard Bright enclosure.
5. Re-fit the sensor retainer having first inspected the sinter to make sure that it has not become contaminated. Contaminated sinters must be replaced as any blockages may prevent gas from reaching the sensor.
6. Ensure the grub-screw is re-secured.
7. Power up the detector and wait till the sensor is stabilized (warm up).
8. Zero and calibrate the new sensor.

**WARNING:** Xgard Bright sensor modules are NOT compatible with Xgard sensor Modules.


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# 4. SPECIFICATIONS

| | |
|---|---|
| **ENCLOSURE MATERIAL** | ADC 12 aluminium alloy or 316 stainless steel |
| **SIZE** | Aluminium Enclosure: 156 x 166 x 109 mm (6.1 x 6.5 x 4.3 inch)<br>Stainless Steel Enclosure: 165 x 166 x 105 mm (6.1 x 6.5 x 4.1 inch) |
| **WEIGHT** | Aluminium enclosure<br>ATEX/IECEx: 1kg (2.2 lbs)<br>UL: 1.5 kg (3.31 lbs)<br><br>Stainless Steel enclosure<br>ATEX/IECEx: 3.1 kg (6.8 lbs)<br>UL: 3.5 kg (7.72 lbs) |
| **CABLE GLAND ENTRY** | 2x M20 or 2x ½" NPT<br>Stopping plug fitted to the left-side entry by default |
| **OPERATING VOLTAGE** | 12V - 30V DC |
| **POWER** | <3W |
| **OUTPUT** | 3-wire 4-20mA (Sink or Source)<br>RS-485 Modbus RTU<br>HART (Optional) |
| **RELAY** | Alarm 1, Alarm 2, Fault (SPDT Contacts. Rated 1A, 24Vdc) |
| **SOUNDER OUT** | High Side Switch (24V nominal, 12V minimum, 700mA max load) |
| **OPERATING TEMPERATURE** | ATEX/IECEx: -40°C to +70°C (-40°F to 158°F)<br>UL: -40°C to +60°C (-40°F to 140°F)<br>Refer to the datasheet for sensor operating temperatures. |
| **OPERATING HUMIDITY** | 0 to 95% RH (non-condensing)<br>Contact Crowcon for sensor operating humidity ranges. |
| **REPEATABILITY** | Repeatability: +/- 2% FSD<br>Zero drift: +/- <2% FSD per year |
| **INGRESS PROTECTION** | IP65/66, EN 60529 |
| **EMC COMPLIANCE** | EN50270 |


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# 5. ACCESSORIES & REPLACEMENT PARTS

| PART NO | DESCRIPTION |
|---|---|
| C01051 | Collector cone |
| C01052 | Spray deflector |
| C01442 | Weatherproof cap |
| C01339 | Flow adaptor |
| C011063 | Sun shield |
| C03005 | Calibration adaptor |
| C01647 | M4 Earth Stud Assembly |
| M04885 | Sensor Seal |
| M03871 | Threaded Hex Spacer 15mm (M4) |
| M03870 | Threaded Hex Spacer 10mm (M4) |
| M03760 | Spare M4 x 8 Posi Pan Head Screw |
| M03810 | M3x6 Grub Screw |
| M02125 | Spare M20 to 1/2" NPT Adapter |
| C03756 | Cro-Mag |
| MIS99033 | Allen Key |
| S015135/S | Sensor Module Flammable 0-100% LEL (MEMS) |
| S015141/S | Sensor Module Flammable 0-100% LEL (MEMS) – UL |
| S015100/S | Sensor Module Methane 0-100% LEL CH4 (Catalytic) |
| S015120/S | Sensor Module LPG 0-100% LEL (Catalytic) |
| S015121/S | Sensor Module Hydrogen 0-100% LEL H2 (Catalytic) |
| S015130/S | Sensor Module Pentane 0-100% LEL C5H12 (Catalytic) |
| S015101/S | Sensor Module Oxygen 0-25% Vol |
| S015142/S | Sensor Module Lead Free O2 Long Life Oxygen 0-25% Vol |
| S015107/S | Sensor Module Hydrogen Sulphide 0-10ppm |
| S015102/S | Sensor Module Hydrogen Sulphide 0-25ppm |
| S015103/S | Sensor Module Hydrogen Sulphide 0-50ppm |
| S015104/S | Sensor Module Hydrogen Sulphide 0-100ppm |
| S015105/S | Sensor Module Hydrogen Sulphide 0-200ppm |


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| PART NO | DESCRIPTION |
|---|---|
| S015109/S | Sensor Module Carbon Monoxide 0-50ppm |
| S015110/S | Sensor Module Carbon Monoxide 0-100ppm |
| S015111/S | Sensor Module Carbon Monoxide 0-200ppm |
| S015112/S | Sensor Module Carbon Monoxide 0-250ppm |
| S015113/S | Sensor Module Carbon Monoxide 0-500ppm |
| S015114/S | Sensor Module Carbon Monoxide 0-1000ppm |
| S015115/S | Sensor Module Carbon Monoxide 0-2000ppm |
| S015119/S | Sensor Module Carbon Dioxide 0-5% vol CO2 (IR) |
| S015118/S | Sensor Module Methane 0-100% LEL CH4 (IR) |
| S015122/S | Sensor Module Pentane 0-100% LEL C5H12 (IR) |
| S015123/S | Sensor Module Propane 0-100% LEL C3H8 (IR) |
| S015128/S | Sensor Module Butane 0-100% LEL C4H10 (IR) |
| S015129/S | Sensor Module LPG 0-100% LEL (IR) |


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# 6. WARRANTY

Equipment leaving our factory are fully tested and calibrated. If within the warranty period of 2 years from despatch, the Xgard Bright transmitter is proved to be defective by reason of faulty workmanship or material, we undertake at our discretion either to repair or replace it free of charge, subject to the conditions below. Sensor module warranty periods are stipulated on technical note GEN067 warranty periods.

**WARRANTY PROCEDURE**

To facilitate efficient processing of any claim, please visit: www.crowcon.com/support/warranty/

You will receive a warranty reference number which must be clearly shown on the address label when the goods are returned. The guarantee will be rendered invalid if the detector is found to have been altered, modified, dismantled, tampered with, or has not used Crowcon spares for replacement parts or has been serviced or repaired by any party not authorised and certified by Crowcon to do so.

For warranty and technical support enquiries please contact:

Telephone: +44 1235 557711

Email: technicalsupport@crowcon.com
warranty@crowcon.com

**WARRANTY DISCLAIMER**

Crowcon accept no liability for consequential or indirect loss or damage howsoever arising (including any loss or damage arising out of the use of the detector) and all liability in respect of any third party is expressly excluded. This warranty does not cover the accuracy of the calibration of the unit or the cosmetic finish of the product. The unit must be maintained in accordance with the instructions in this manual.

The warranty on replacement consumable items supplied under warranty to replace faulty items, will be limited to the unexpired warranty of the original supplied item.

Crowcon reserves the right to determine a reduced warranty period, or decline a warranty period for any sensor supplied for use in an environment or for an application known to carry risk of degradation or damage to the sensor.

Our liability in respect of defective equipment shall be limited to the obligations set out in the guarantee and any extended warranty, condition or statement, express or implied statutory or otherwise as to the merchantable quality of our equipment or its fitness for any particular purpose is excluded except as prohibited by statute. This guarantee shall not affect a customer's statutory rights.

Crowcon reserves the right to apply a handling and carriage charge whereby units returned as faulty, are found to require only normal calibration or servicing, which the customer then declines to proceed with.

**PRODUCT USE:**

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# CONTACTS

## UK HEAD OFFICE

172 Brook Drive, Milton Park, Abingdon, Oxfordshire, OX14 4SD
T: +44 (0) 01235 557700
Crowcon Detection Instruments Ltd.

## NORTH AMERICA OFFICE

5690 Eighteen Mile Road, Sterling Heights, MI 48314, USA
T: +1 (859) 957 1039
Crowcon Detection Instruments Ltd. (USA Office)

## MIDDLE EAST & INDIA OFFICE

Unit B4, Light Industrial Units, Phase 5 – Dubai Silicon Oasis – Industrial Area – Dubai – United Arab Emirates.

T: +971 (0) 0662 0105

Crowcon Detection Instruments Ltd. (Middle East & India Office)

## SINGAPORE OFFICE

Block 194 Pandan Loop, # 06-20 Pantech Business Hub, Singapore, 128383

T: +65 6745 2936

Crowcon Detection Instruments Ltd. (Singapore Office)

## CHINA OFFICE

Floor 3, Building 7, No.156, 4th Jinghai Rd, BDA, Beijing, P.R. China. 101111
T: +86 (0)10 6787 0335
crowcon.com.cn

Crowcon Detection Instruments Limited Company (China Office)

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**For more information:**
t: +44 (0)1235 557700
e: info@crowcon.com
w: www.crowcon.com

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